

Probability & Statistics

BCA — Semester III — 2023 — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

11. Discuss the role of statistics in computer application.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Define Statistics? The following of telephone calls received at an exchange for 200 successive one-minute intervals are given below.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

13. Define correlation. Calculate and analyze the correlation coefficient between the number of study hours and the number of sleeping hours of different students.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

14. Define regression. The following table gives information on ages and cholesterol levels for sample of 10 men.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

15. Define binomial distribution. During one stage in the manufacture of integrated circuit chips, a coating must be applied. If 70% of chips received a thick enough coating, find the probability that among 15 chips: i) at least 12 will have thick enough coatings ii) exactly 10 will have thick enough coatings

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

16. In measuring reaction time, a psychologist estimates that the standard deviation is 0.95 sec, how large a sample of measurements must be taken in order to be 95% confident that the error his estimate of mean will not exceed 0.01 second?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

17. Define sampling. A population consists of the four numbers 1, 3, 4, 8.(1) Write down all possible sample size of two without replacement.(ii) Show that sample mean is an unbiased estimate of population mean.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions[2x10=20]

18. An analysis of monthly wages paid to workers in two firms A and B belonging to the same industry gives the following results:

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

19. The length of human pregnancies from conception to birth approximates a normal distribution with a mean of 266 days and a standard deviation of 16 days.(i) What probability of all pregnancies will last between 240 and 270 days?(ii) What probability of all pregnancies will last more than 275 days?(iii) What probability of all pregnancies will last less than 272 days?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

20. Define ANOVA. Given data, using one-way ANOVA test the hypothesis that means of three samples are equal

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.